

Cognitive Regularities in Creative Activity

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Theories of creativity have not traditionally considered whether novel ideas or inventive behaviors can result from regularities in the cognitive processes responsible for such activities. Most of these traditional theories are based on the evaluation of products as meeting (or not) some abstract metric of creative output. However, cognitive theories of creativity can be proposed in which creative activity is a function of more traditional cognitive processes that are not unique to inventive behaviors. The purpose of this article is to review the cognitive regularities of creative activity and organize the research on this topic into a framework that might be useful in understanding and extending investigations directed at studying creativity. To these ends, cognitive processes underlying generation, synthesis, and selection of information in creative activities are delineated.

Creative activity seems almost magical in its power to fascinate us. (Cari R. Hausman [1984], *A Discourse on Novelty and Creation*, p. 1)

The notion that creative behaviors represent extraordinary ability has been a popular one both in the past and in the present. For example, the ancient Greeks attributed human creativity to the intervention of the muses, whereas both Plato and Popper espoused the view that creativity was mystic and not necessarily rational. Such views have persisted because creative behaviors have eluded a satisfying, concrete definition. In support of this position, little agreement exists among academic disciplines as to the central attributes of creativity (Sternberg, 1985). For example, Sternberg reported that physics professors rated inventiveness and orderliness as the central attributes of creativity, whereas professors of art reported that these attributes were imagination and originality. Because of the variability of these opinions, empirical research on creativity has not converged on rules or definitions that identify performance as creative or not. Unfortu-

nately, there are as many research-based definitions of creativity as there are approaches to studying the topic (see Boden, 1994; Finke, Ward, & Smith, 1992; Sternberg & Lubart, 1996). Common approaches have included psychometric evaluations (e.g., Torrance, 1974, 1988), assessment of social and personality variables (Amabile, 1996; Eysenck, 1993), examination of cognitive processes (Boden, 1991; Finke et al., 1992; Mednick, 1962), and the confluence of several of these approaches (Gruber, 1988; Sternberg & Lubart, 1996).

In our opinion, a common impediment in most of these approaches is the pervasive undercurrent that creative behavior is somehow the product of genius or unusual talent and not the product of more general human abilities that might be possessed by many. As a consequence, the dominant approaches to studying creativity have sought to define what special talents or abilities are possessed by creative people (Weisberg, 1993). These approaches have suggested that creativity goes beyond the normal range of human ability and, therefore, normal cognitive processing. However, as Weisberg (1993, p. 10) stated,

The creative achievement can be extraordinary because of the effect it produces, rather than because of the way in which it was brought about. As a rough analogy, the straw that breaks the camel's back is not extraordinary qua straw, but is extraordinary because of the effect it has on the camel.

This observation is exceptionally important because it highlights the fact that a creative

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achievement can occur through more mundane means but nevertheless still leave a substantial impact on those who encounter it. By Weisberg's definition, the cognitive or generative processes through which a creative product is generated have been viewed as less important to its birth than is the ultimate impact it may have. By downplaying the mental activities involved in the generation of some types of creative achievement, Weisberg's observation is consistent with the vast majority of definitions that have been offered for creativity. These definitions are largely based on criteria that are extraneous to the cognitive processing that gives rise to the product being evaluated, such as the sociocultural evaluation of the finished product or its aesthetic or functional appeal. Because such evaluations can change over time as societal norms evolve, they are unlikely to ever provide a consistent standard on which to evaluate behaviors as creative or on which to base future intellectual inquiry into the complex topic of creativity.

Approaches that evaluate a final product against standards that may be fleeting or otherwise contextually bound are defined as product-centered approaches to evaluating creativity. Although such definitions will always constitute some aspect of what observers judge to be creative versus not creative, a central theme of our arguments is that another fruitful approach for studying creativity would be to understand and to specify common underlying cognitive processes that give rise to both more creative and less creative products. According to this approach, it is possible to understand important aspects of creativity by focusing on how information is generated and processed in service of generating novel products. Thus, in taking Weisberg's analogy one step further back, our thesis here is that it is worthwhile to understand the origin of the straw separate from its ultimate effect on the camel. By origin, we mean universal cognitive processes that are possessed by most people (rather than few) and that support the generative activities ultimately labeled creative versus more mundane.

By way of overview, our central argument is that creative processes involve many of the same underlying cognitive processes that are present in more mundane, everyday activities. This is not a novel argument in and of itself. Other researchers have explored the cognitive

processes of creativity and offered cognitive models of creativity (e.g., Finke et al., 1992; Mednick, 1962; Runco & Chand, 1995). The point of departure for the present work is in providing an organization of basic cognitive processes in a way that allows for creative activity, thus elaborating on previous work in creative research and cognitive research. In doing so, we attempt to demonstrate that universal cognitive processes form the foundation of creativity and that these mechanisms are possessed by the majority rather than the few. Ultimately, we argue that there are individual differences in evaluating what these processes offer up as novel or original but that these processes do not functionally differ between the genius and those who appear (*prima facie*) less gifted.

First, we provide a brief overview of the similarities between the processing involved in creative versus noncreative tasks. In doing so, we summarize two basic theories of creative behavior and the similarities between them. Second, we develop a framework that was designed to subsume the cognitive activities that are most central to generative activities whose products will ultimately be labeled creative or not. Our framework is composed of two central activities defined shortly. These activities are elaborated in turn to specify the central cognitive mechanisms contributing to each. Some consideration is given to how individual differences in these processes might lead to more versus less creative innovations. Finally, we conclude with suggestions on how to integrate the framework into other approaches to studying creativity.

Cognitive Universals in Creative Activity

The cognitive processes that support creative activities have not received sufficient concentrated research efforts (Sternberg & Lubart, 1996). Nevertheless, some researchers have demonstrated that creative activity does involve very regular, cognitive processes (e.g., Finke et al., 1992). Evidence for this claim comes from demonstrations that people use information in similar ways to perform both creative and noncreative laboratory tasks (Marsh, Landau, & Hicks, 1996; Smith, Ward, & Schumacher, 1993; Ward, 1994). For example, in noncreative categorization tasks, people use highly character-

istic features of exemplars as their basis for judgments about category membership, such as deciding whether an ostrich, a robin, or a bat is a bird (Tversky & Hemenway, 1984). For the category bird, such features include size, presence of feathers, and the attribute of flying. Likewise, in creative tasks, Ward (1994) has demonstrated that people use highly characteristic features of specific categories as a basis for generating the components of novel drawings of space creatures or toys for children. The important point is that the same attributes that are central to noncreative tasks also appear to be central to more creative endeavors.

Other types of information that are critical to both noncreative categorization and creation of novel designs (e.g., designing bicycle racks; Jansson & Smith, 1991) include the appearance of functional features and correlated attributes (Ward, 1994, 1995). For example, a functional feature of a novel animal would be sense organs for obtaining information, whereas such a feature of a toy might involve movement or the attribute of exercise. These attributes mirror those observed on Earth animals or contemporary children's toys. In terms of correlated attributes, birds usually have beaks and wings, and this co-occurrence appears in noncreative attribute listing tasks as well as in creative, novel drawings. Still other similarities have been noted. For instance, Smith et al. (1993) and Marsh et al. (1996) found that providing examples in creative drawing tasks led people to copy surface features of those examples, much like surface details of previously solved problems influence noncreative problem solving by analogy (Gick & Holyoak, 1980; see also Ross, Ryan, & Tenpenny, 1989). Such examples are common and are elaborated further in subsequent sections.

Findings such as these led Finke et al. (1992) to devise the Geneplore model of creative behavior. This model was intended to delineate the cognitive processes involved in domain-independent creative activities. According to the model, creative activity is the process of generating, refining, and then regenerating mental representations in service of task demands and goals. In Geneplore, there are two major types of cognitive processes used in creative activity: those involved in generation and those involved in exploration. Generative processes construct initial mental representa-

tions in incomplete forms known as preinventive structures. The properties of preinventive structures promote further refinement through the exploration processes that operate on them. In the exploration phases of creative behavior, the preinventive structures are interpreted in meaningful ways and are modified to satisfy the specific goals of the task at hand. For example, when individuals generate a novel space creature, they often start by drawing a body (Finke et al., 1992). This preinventive structure is then refined by mentally exploring the specific form that the body will take. Presumably, memory is searched for different body forms such as that of a dog, a fish, a snake, a bird, and so forth. Task constraints such as generating a novel inhabitant of a planet predominantly covered with molten lava (e.g., Ward, 1994) will influence the choice and form that the exploration processes take (in this specific case, participants often create birdlike creatures to avoid the planet surface).

This model represents a significant contribution of cognitive psychology to understanding creativity. Perhaps its largest contribution is that it directs research on creativity toward critically examining the cognitive processes by which information is generated in novel situations. In terms of Weisberg's (1993) example, the model shifts the research focus back one phase in asking whence the straw originated that ultimately broke the camel's back. In so doing, our thesis is that many aspects of the processes that precede or otherwise contribute to a final product can be defined as analogous (or identical) to, say, standard memory retrieval processes that have been studied for many years in other, more traditional contexts. As a consequence, the principles from this tradition can be fruitfully applied to understanding creative endeavors. In addition, the Geneplore model specifies that there are constraints on the form and function of creative products that arise from the specific nature of the task or the generative processes involved. These constraints demonstrate additional regularities in cognitive processing insofar as they focus generative processing on retaining certain domains of information as well as discarding or abandoning others.

The Geneplore model of creative behavior shares some similarities with other theories of creative behavior. For example, Runco and Chand's (1995) model is a two-tier hierarchical

model in which primary processes (e.g., ideation and evaluation) interact with secondary processes (e.g., motivation and knowledge) to produce novel products. The two critical features of Runco and Chand's model are that it stresses the importance of the cognitive underpinnings of creativity and that those cognitive processes must interact to produce truly novel entities. The same is true in the Geneplore model, in which preinventive structures that are produced by generation processes must be refined by exploratory processes that modify these incipient structures to meet specific goals or task demands. Therefore, although Finke et al.'s (1992) and Runco and Chand's models differ in their labeling of certain creative processes and in whether these processes are hierarchically organized, there is much similarity in the iterative nature and interaction between the cognitive components necessary to move from a rather ill-defined goal to a more or less well-specified novel product. The framework that we have developed borrows somewhat from the basic overarching processes specified in these models but attempts to delineate the more specific, detailed mechanisms that support these broader, overarching processes. We know of no model that has hitherto attempted to provide a specification of the underlying detailed components of these broader processes, although Finke et al.'s and Runco and Chand's models are certainly the precursors to the current endeavor.

Creative Versus Noncreative Tasks

Because we argue here quite strongly that many of the cognitive processes that support creative behavior also support more mundane activities, it is important that we first differentiate between creative and noncreative task demands. Although cognitive processes and mechanisms may be similar, information is not necessarily used in the same way in response to the two different task demands. In noncreative tasks, information is retrieved from memory on the basis of its association with a particular encoding context or particular learning episode (Hintzman, 1986; Jacoby, Levy, & Steinbach, 1992; Kolers & Roediger, 1984; Tulving, 1962). By nature, explicit memory tasks such as recognition or categorization judgments demand that elements of the encoding context be

reconstructed. By contrast, creative tasks are less well defined with regard to what memory structures are relevant to the task (Finke et al., 1992). Although creative tasks demand that information from certain domains be used, fewer constraints are placed on the range or the variability of the information that is task relevant. Therefore, although memory retrieval of some sort is involved in both creative and noncreative endeavors, the hallmark of creativity may be how widely memory is searched. Stated differently, the "relevant search context" may be wider or less specifically defined in creative tasks than it is in everyday tasks. There is growing evidence to suggest that creative generation uses more global knowledge structures (Ward, 1994; Ward & Sifonis, 1997) and consequently less specific information to search for relevant pieces of information that can be used to meet task demands (Marsh, Bink, & Hicks, 1999). Thus, the distinction to be made between creative and noncreative cognition lies in what information people consider applicable to the goals of the task. At the extreme, what one chooses as applicable for a given task may be the only difference between creative and noncreative cognitive tasks.

The choice of task-relevant information necessarily requires processing that augments memory retrieval (i.e., initial generation processes). Such additional memory processes have been well specified in theories of noncreative cognition. For example, theories of both recall (J. R. Anderson & Bower, 1972) and recognition memory (Mandler, 1980) generally posit two-process mechanisms. The same is largely true of theories concerning how people determine the origin of information (i.e., source monitoring; Johnson, Hashtroudi, & Lindsay, 1993) or how people make situational frequency judgments (Begg, Maxwell, Mitterer, & Harris, 1986). One set of processes retrieves relevant information, whereas another determines whether the information is suitable to the tasks and goals at hand. Such two-process theories have been known as generate-and-edit models of memory (Zechmeister & Nyberg, 1982). On the basis of notions such as these, we elaborate a framework involving generation processes followed by iterative selection mechanisms. In the generation phase, relevant knowledge structures are recruited from related domains as a function of broadly defined memorial searches. During this

phase, pieces of information are temporarily synthesized together to form potentially novel combinations. In the selection phase, which of these combinations are to be retained or discarded is determined with some subset usually constituting memorial cues to further generation, synthesis, and continued selection.

The components of this framework generally define how information is generated in service of creative tasks and therefore can be used to organize the relevant research that is germane to the cognitive underpinnings of creative behavior. The framework represents a generalization of the Geneplore model and Runco and Chand's (1995) model, and it also borrows from traditional generate-and-edit models of memory retrieval. According to the framework, creative cognition consists of two general interacting processes. These processes of selection and generation are defined broadly in the next section, and then their component cognitive processes are elaborated in detail in subsequent sections.

Overview of Generation and Selection Processes

Generation processes are largely analogous to memory retrieval processes in noncreative cognition. However, these processes not only recruit task-relevant information but also begin to synthesize that information. Interim synthesis allows partially retrieved information to be combined and to then serve as a retrieval cue for the iterative probing of memory for further relevant or novel pieces of information. Although this characterization seems conscious and deliberate, these generative processes actually exist on a continuum between the use of direct information (i.e., consciously directed retrieval) and the use of more indirect information (i.e., unconsciously retrieved). The distinction between direct and indirect uses of information is used here in the traditional sense that it is used in the cognitive literature (see Schacter, 1987; see also Jacoby, 1991). The use of direct information in creative tasks results from retrieving specific information related to the domain in question or relevant to the explicit constraints of the task (i.e., goals). This information is not necessarily associated with a particular episode (Roediger, Weldon, & Chellis, 1989) but is simply the result of intentional,

goal-directed processing (Jacoby, 1991). For example, in reasoning by analogy, knowledge acquired in a particular context is applied more or less directly in a new context (see Gick & Holyoak, 1980). The use of directly retrieved information may also take the form of generalizing from a particular category, mental model, or other similar higher order knowledge structure. An example of this type of direct process is illustrated by Ward's (1994) theory of structured imagination, in which people sometimes consciously base their novel creations on existing category knowledge (e.g., using Earth animals as a basis for novel space creatures).

Indirect uses of knowledge in creative tasks are governed by the mechanisms of implicit memory. In these cases, information from prior experience is retrieved and used in the solution to a task outside of one's conscious awareness. Because creative tasks can be defined as having ill-defined end states, the search for specific information may result in the use of available information. However, demands of creative tasks may still influence the type of information that is made available. Thus, the final form of a creative product may represent the interaction between information familiar to the individual and specific task demands.

Synthesis processes operate in intimate conjunction with generation processes. Synthesis involves combining pieces of information that have not hitherto been associated with one another to yield a separate piece of new information. A simple example of synthesis is the combination of forms in mental imagery. Existing patterns or shapes may be combined to form a new mental object (Kosslyn, 1987). This new object is then able to be stored and retrieved independent of its components (Thompson & Klatzky, 1978). Perhaps the best example of synthesis is conceptual combinations of information. These combinations will yield products that have properties independent of their elemental entities. For example, milk sand and cactus pig are random combinations of words that people can readily understand as conceptual combinations, as demonstrated by their providing meaningful definitions as "white sand" and "prickly pig," respectively. The growing literature on conceptual combination suggests that individual concepts such as wooden and spoon that are placed together as a single entity of wooden spoon often result in emergent features

of the conceptual combination that are not listed separately (e.g., Gagne & Shoben, 1997; Shoben & Gagne, 1997; Wisniewski, 1997). One emergent property of combining wooden and spoon is that wooden spoons are large. Even though conceptual combinations are most prevalent with unusual combinations (Hampton, 1997), the important point is that processes that synthesize, or bring together, novel entities often instill in the combinations properties that the elemental entities individually do not possess. People do not list talking as an attribute of pets or birds separately but will often do so when asked to list the attributes of pet birds (e.g., Hampton, 1997). Thus, the same must be true when people are engaged in creative endeavors. The melding and combining processes associated with synthesis bring together novel features that combine to form a product greater than the sum of its parts. Although generation and synthesis are separate cognitive processes, they have been subsumed together in this framework under the label of generation because of the highly iterative nature in which they must work.

Whereas generation processes produce candidate information that can be used in creative endeavors, selection processes form that information into a final product. Selection processes are higher level processes that determine what information will be used and how it will be incorporated to meet the goals of the task. One mechanism of selection processes is choosing a subset of information from generation processes that will be retained or discarded. In this case, selection involves determining the suitability of generated information for the task. Selection processes can retain or reject elemental representations as well as retain or discard combinations that have been generated during synthesis.

As an example of how selection processes work, consider the task of designing a novel toy (e.g., Smith et al., 1993). One may briefly consider the category of gemstones and decide to design new plastic "jewelry" for young girls whose pieces are interchangeable and can be strung together to make bracelets or necklaces. In the generation process, other categories may have been considered (e.g., tools) but rejected by selection processes as ill suited or not amenable to the generative task at hand. Selection processes can also reject novel combinations of ideas, such as the plastic jewelry, as

either not novel (which the jewelry is not) or an unsuitable solution (because it does not fit the constraints and goals of the task at hand; e.g., perhaps the novel toy should be suitable for both young boys and young girls). The important point is that selection processes guide the selection of information to be combined in novel ways and provide a check on the current form of the novel combination for its appropriateness to the creative goal at hand. Although people are quite adept at generation and synthesis processes (e.g., Ward, 1994), the selection process appears to be somewhat more fragile and more prone to errors than is the generation process. For example, Marsh and his colleagues have repeatedly shown that people do not monitor their "novel" ideas sufficiently to avoid unconscious plagiarism of other people's novel ideas (e.g., Marsh et al., 1996; Marsh, Landau, & Hicks, 1997).

At this stage of development, the framework makes no assumptions as to whether generation and selection occur in serial or parallel. Selection processes may occur on-line, spontaneously, or serially depending on the particular task at hand and the nature of the processes involved. However, the relationship between generation and selection is important in understanding creative cognition. For example, consider how cognitive agendas might influence generation processes. Perkins (1988) offered a theory of creativity whereby people "hark back" to previous episodes to generate new information. Accordingly, such an agenda would necessitate the use of explicit retrieval processes to generate information. Although Perkins has depicted the creative process as more explicit, as argued earlier, creative endeavors are likely to be influenced by unconscious processes as well (e.g., Marsh, Ward, & Landau, 1998). In the novel drawing paradigm, for example, it has been demonstrated that the features participants include in their designs can be influenced by an "unrelated" task (Marsh et al., 1999). The same is true when participants plagiarize other people's ideas of ways to reduce traffic accidents or when they generate novel nonwords. These results clearly implicate unconscious influences as playing an extraordinarily important role in creative tasks. In this case, selection processes may work in parallel with (or perhaps after) generation processes to determine the suitability

of interim products to the ultimate requirements of the task being undertaken.

Having now outlined the general mechanisms of the framework, we intend, in the following sections, to describe in further detail the cognitive processes associated with creative activity. The treatment of these issues is not intended as an entirely exhaustive review of the relevant literature. Rather, we present examples from the literature that are important in broadly defining potential universal mechanisms that support creativity. In the section on generation, the goal is to specify common cognitive processes that are involved in creative activities, drawing parallels between the creative and the noncreative cognition literatures. The selection section provides additional support of the general mechanisms previously described. Therefore, the goal of these sections is to offer a more complete explanation of how cognitive agendas and selection processes contribute to creative activity. In addition, each section addresses how generation and selection can explain why some people are labeled more creative than others in terms of their use of particular cognitive processes.

Generation and Synthesis Processes

Perhaps the most regular finding of creative cognition is the influence of prior exposure on the final form of the creative product. Prior exposure to information facilitates a range of noncreative memory tasks from word fragment completion to problem solving. These tasks are facilitated through making information available both within and outside of conscious awareness, through abstraction from existing knowledge structures, and through transfer of information across tasks. Thus, prior exposure serves as a basis for direct (conscious) and indirect (unconscious) retrieval of information associated with any task (J. R. Anderson & Bower, 1973; Bower, 1996). As argued here, such information can serve as a basis for generation as well. In creative tasks, the most important influence will be on generation processes; to a slightly lesser extent, there will be an influence on synthesis processes. In this section, we first discuss the more unconscious influences in creative tasks and their noncreative counterparts as well. Then we consider the amalgam of conscious and unconscious influ-

ences, followed by the more traditional conscious or explicit cognitive processes that contribute to creative behavior.

Unconscious Influence of Past Experience

In the traditional sense, priming generally refers to the increased probability of responding in particular tasks with recently encountered information as compared with less recent information. Of course, the defining characteristic of pure forms of priming is that they are unconscious processes. Many theories exist for the various forms of priming that have been studied in the cognitive literature (see Bower, 1996; see also Tenpenny, 1995). However, Roediger and Challis (1992) have demonstrated that broad generalities exist across different forms of priming. The majority of the priming literature analyzes implicit memory in tasks such as perceptual identification, word stem completion, and word fragment completion. Roediger and Challis have extended these findings to explicit tests of memory such as free recall as well. For example, their participants studied a list of words for a later unspecified memory test. The study list consisted of target words such as *tornado*, words visually similar to the target (e.g., *Toronto*), and three types of conceptually similar words. The conceptually similar words included a synonym of the target (e.g., *twister*), an associate of the target (e.g., *basement*), and a category coordinate of the target (e.g., *typhoon*). When participants were asked to perform a word fragment completion task for target words or to take a free-recall test, the results depended on the type of prior exposure to list items. Word fragment completion was affected by the mere presence of items (e.g., visually similar words), whereas free recall was differentially sensitive to the type of conceptual relation between studied items (e.g., category coordinate vs. synonym). The important point of this study for the present purposes is that the influence of prior exposure is a function not only of how information was encountered earlier but also of the nature of the task demands that recruit that information. In this case, noncreative cognitive tasks are differentially sensitive to prior exposure and task demands, and we demonstrate here that so too are more creative tasks.

The Roediger and Challis (1992) study should

also highlight the fact that generation processes cannot be defined as recruiting information strictly by implicit or explicit means; rather, it is more likely to be the result of some combination of the two. Therefore, the generation of relevant information in creative tasks can result from both indirect and direct routes. For example, Smith et al. (1993) had participants either design new toys or draw novel space creatures after showing half of the participants relevant examples. Unbeknownst to the participants, the examples of each category shared three common features (e.g., four legs, tails, and antennae on space creatures). Although the participants were admonished to not copy any aspect of the examples, the group of participants exposed to them included those features in their own novel designs about 12% more often than did participants who saw no examples at all. This effect persisted over a short (23-min) delay (Experiment 2). Marsh et al. (1996) extended this finding by demonstrating that different levels of prior exposure would affect generative designs. Their study varied the number of examples (0, 3, 6, or 9) presented to participants and showed that the greater the number of examples shown, the higher the probability of including the common features. Together, these two studies represent evidence of more direct and less direct priming from previous exposure. On the one hand, if the examples were able to be avoided according to the admonition instructions, there should have been little conformity to them (an implicit influence). On the other hand, people copied the forbidden features more often than was the case in control conditions, suggesting a more explicit influence.

As described earlier, Marsh et al. (1999) also demonstrated an even more indirect (i.e., unconscious) form of priming. In this case, the creative drawing task was preceded by an "unrelated" sentence construction task that provided a mild hostility prime. This prime resulted in increased inclusion of hostile and negative features in people's novel creations. Interestingly, the level of indirect priming was equivalent to the level of direct priming arising from seeing examples of creatures that actually contained hostile features. Therefore, this study illustrates the influence of priming in a creative paradigm that can result in similar (or different) outcomes depending on what material is studied, much as Roediger and Challis (1992) had

shown in a noncreative paradigm with different types of study materials.

Influence of Knowledge Structures

Although these implicit influences on creative behavior may seem obvious when described, the majority of research on creative and noncreative tasks concerns the influence of well-established knowledge structures, as opposed to more recently encountered information. Our previous focus in describing the implicit influences on creative tasks was not intended to deny that long-standing knowledge structures influence creative and noncreative cognition as well. After all, knowledge structures provide organization for the retrieval and generation of information, and their strong impact on cognitive processes can hardly be denied. However, the important point is that accessing such structures is not necessarily the consequence of direct retrieval by way of conscious strategy. These structures can be recruited to support task demands in a more automatic form, say, based on the semantic context of the generative task being undertaken. In one classic series of studies, Bransford and Johnson (1972, 1973) demonstrated the effects of prior knowledge on the ability to comprehend and retrieve ambiguous information. They asked people to read passages composed of related yet ambiguous sentences. When a topic for the passage was presented before the passages were read, people were better able to comprehend the sentences and recall information from the sentences. The topic (e.g., "washing clothes") provided a schema to support learning the information. Thus, one's prior knowledge of the schema helped to organize the ambiguous sentences as they were encoded.

Perhaps more important for the current purposes, the schema could also serve as a basis for retrieval of information. In Bransford and Johnson's later study (1973), people were provided with one of two different topics for the same passage. People recalled information most consistent with the specific topic that was provided to them (e.g., watching a parade vs. watching a moon landing). A further illustration of this mechanism comes from the now classic "burglar versus home buyer" experiments (e.g., R. C. Anderson & Pichert, 1978). In these experiments, people read a story with the perspective of either a burglar or a home buyer

in mind. In general, people recalled information consistent with the schema used to learn the information. However, when given the other perspective at test, people recalled more information consistent with the test schema than with the schema present at learning. Taken together, these results suggest that people use relevant knowledge structures as a starting point for memory retrieval regardless of how information was originally acquired.

The obvious implication for creative tasks is that the semantic context of a particular task will influence what prior knowledge is brought to bear. Ward's (1994) theory of structured imagination is based on the very tenet that creative imagination is influenced by semantic knowledge structures such as preexisting categories and mental schemas (see Finke et al., 1992, chap. 6). In the drawing task to which we have referred throughout this article, Ward found that drawings of aliens consistently contained attributes similar to Earth animals when no other information was provided to participants. For example, people's novel creations were uniformly bilaterally symmetric, contained sense organs for obtaining information, and had a means of locomotion. One interesting aspect of these results is that when participants were asked to list explicitly the features of Earth animals, they did not list bilateral symmetry. Thus, the creative process is sensitive to features that are generally not available to explicit cognitive processes, at least as measured by attribute listing tasks.

Ward (1994) also found that abstractions from the category of Earth animals were very regular depending on the constraints provided by task instructions. For example, novel drawings contained more features typical of Earth animals when participants were asked to draw aliens to inhabit a planet similar to Earth as compared with a planet that was very different. In addition, participants drew features that tended to be correlated with one another in nature. That is, feathers and beaks were included in drawings of creatures that participants were also told had wings. This finding suggests very strongly that creative imagination draws on information in existing knowledge structures as a relevant starting point for generating novel entities. This assertion parallels the finding of Bransford and Johnson (1972, 1973) and R. C. Anderson and Pichert (1978) that such information also serves

as a starting point for noncreative memory retrieval.

Prior Knowledge and Synthesis in Generation Processes

The very fact that participants tend to use correlated features is as informative about generating relevant information in creative tasks as about how that information is combined. As we argued earlier, generating relevant pieces of knowledge is hardly creative unless those components can be combined in novel ways. Therefore, combination or synthesis constitutes an exceptionally important part of creative behavior. If participants generate features of animals when drawing novel space creatures, then how those features are combined will influence how "creative" the final product is perceived to be. The fact that a climate of molten lava results in winged creatures and one of gaseous upper atmospheres results in land dwellers suggests that not only are features synthesized together in very regular ways, but these features are also synthesized with the constraints of the task.

Marsh et al. (1996) included animal features in examples that did not co-occur in nature (e.g., claws and antennae). Participants who saw such examples did not use those features in novel drawings, whereas the combination of naturally co-occurring features (e.g., four legs and a tail) had a dramatic impact on performance. Results such as these suggest that although people may attempt to recombine information in novel ways, the synthesis of components that are elicited during generation tends to conform to preexisting knowledge structures. Perhaps it is for this reason that participants are reluctant to deviate from novel creations that do not possess bilateral symmetry, that contain odd means of locomotion, or that include strange mechanisms for obtaining information from the environment. Moreover, even when participants do include such extraordinarily odd attributes, those attributes almost invariably preserve the same functions performed through analogous features on Earth animals (Bredart, Ward, & Marczewski, 1998).

This tendency of synthesis processes to recombine features in a manner consistent with preexisting knowledge structures is not unique to creative behavior. It is quite common in

noncreative cognitive tasks as well. As an example, people have extreme difficulty in learning categories whose attributes conflict with real-world knowledge (e.g., Murphy & Allopenna, 1994; Spalding & Murphy, 1999). When the defining attributes of a category are "sharp" and "used for smoothing wallpaper," for instance, category acquisition proceeds quite slowly. Such attributes are inconsistent with real-world knowledge (i.e., preexisting knowledge), and therefore novel learning is hampered. However, when "sharp" and "used for killing bugs" constitute the defining attributes of a category, acquisition proceeds much more quickly. Therefore, in both creative and noncreative tasks, the synthesis of feature constellations will be facilitated if those constellations are consistent with prior experience.

A more direct and perhaps more conscious form of this phenomenon is the use of prior knowledge in analogical transfer. When analogy is used successfully, specific information from one context is used to solve a novel problem in another context. Obviously, analogy is used in problem-solving tasks in which the solution to one problem is used as the basis for the solution to another problem. For example, Gick and Holyoak (1980, 1983) presented people with a logical story problem that few people could solve initially. However, when people were presented with a different cover story of a functionally equivalent problem and the original solution as a hint, performance improved dramatically. The use of analogy to solve problems successfully depends critically on the similarity of the prior solution and the current problem state (Ross, 1987). Such laboratory problems represent less creative endeavors, but the same is nevertheless true in more creative activity (Boden, 1994). For example, Perkins (1988) offered a theory of creative cognition fundamentally based on "harking back" to old ideas. He suggested that when people engage in creative tasks, they actually attempt to recall explicitly prior instances of the task and other relevant information.

The drawing task described throughout can be used to illustrate this mechanism. When people begin to draw space creatures, they first "hark back" to previously encountered space aliens as depicted in movies, magazines, science fiction novels, and television programs. With this information in mind, an attempt is then made to

recombine the features from these previous examples, along with other information (e.g., features of Earth animals), to create a novel design. In these cases, the use of analogous features is a direct strategy for meeting task demands. Under such circumstances, generative processes are more explicit, and it is not very surprising that the synthesis processes used to combine components would conform more explicitly to existing knowledge structures.

Failure to Use Prior Knowledge

The preceding discussion has documented some of the basic mechanisms for generation believed to be used in creative tasks. The discussion has focused on the impact of prior knowledge, both direct and indirect, on the generation of creative products. In this discussion, parallels were drawn between the processes observed in creative and noncreative tasks. In general, the evidence seems to suggest that the processes of generation and synthesis involve the same cognitive processes as those observed in noncreative cognition. However, as in noncreative cognition, the impact of available information is not always perfectly predictable. There are well-documented instances in which prior knowledge is not used in service of tasks even though it would be applicable. For example, Weisberg, DiCamillo, and Phillips (1978) gave participants paired-associates training before they attempted to solve problems. The paired associates contained clues that would aid in problem solving. However, Weisberg et al. found that people did not spontaneously use the "clues" in generating problem solutions (i.e., a failure of implicit priming). In fact, solutions were facilitated by the prior exposure only when specific "hints" were given to remember the paired-associates training (similar to the discussion of analogy).

Weisberg et al. (1978) concluded that their participants were focused on searching long-standing, problem-specific memory rather than more recent, local memory. That is, they used general knowledge of the problem (or elements of the problem) and not necessarily recent experience to facilitate their efforts on the task. Perfetto, Bransford, and Franks (1983) obtained a similar result using different tasks, which lends some support for Weisberg et al.'s explanation, and they offered an additional

interpretation for this phenomenon. Perfetto et al. not only suggested that the search of memory for solutions can be problem specific but also that initial retrieval attempts may guide subsequent attempts. They argued that if recent information is understood to be relevant to problem solving, it will tend to be used; conversely, if this information is not seen as relevant, only information associated with the initial retrieval attempts will be used.

This interpretation dovetails well with our thesis that generation and synthesis processes are intimately related and influence one another at various stages in the creative process. Reinterpreting Perfetto et al.'s (1983) thesis in terms of the current framework, initial generation processes attempt to produce relevant pieces of information that are then synthesized into potentially novel combinations. If those combinations appear fruitfully to move an individual closer to task goals and meeting relevant constraints, then the interim structures can serve as additional retrieval cues for generating even more related information out of memory. Our proposition that interim combinations of elemental pieces of information can guide further retrieval attempts is not at all unique to creative endeavors. Rather, this proposition is grounded squarely on analogous findings with noncreative cognitive measures. For example, Ratcliff and McKoon (1981) have proposed that semantic priming effects in lexical decision (i.e., word-nonword judgments) can be explained by their compound-cue theory of human memory. In such tasks, words like *nurse* will greatly increase reaction times (i.e., prime) to related words like *doctor* if they are judged in fairly close succession to one another. However, inserting even one or two unrelated words removes this semantic priming effect. Ratcliff and McKoon have argued that the compound cue of *nurse* as the context will facilitate the memory search for *doctor*, whereas *nurse* combined with an unrelated word (e.g., *tape*) serves as a diluted compound cue inhibiting the rapid convergence on *doctor*. Therefore, Perfetto et al.'s thesis of initial generation processes influencing later retrieval attempts is entirely consistent with such compound-cue theories of memory retrieval.

The failure to use available information consistently represents an important issue in research on creativity, and it highlights the fact

there is some individual variability in cognitive processing. This variability may constitute what distinguishes more creative behavior from less creative behavior. In so stating, we are not retreating to the position that creativity, per se, is an individual difference. On the contrary, variability in noncreative cognitive processes can be seen within individuals across even short periods of time. For example, people are known to categorize objects such as a tomato as a fruit on one occasion only to retreat and label it a vegetable minutes later (e.g., Rosch, 1978). Because there is variability in the underlying cognitive processes in noncreative tasks, one should also assume that the same is true in more creative endeavors. Obviously, some individuals may be more consistent than others. According to the present framework, the occurrence of more creative versus less creative outcomes on different occasions for a given individual is likely to be a function of what elemental pieces of information are generated and how synthesis processes combine those pieces. Moreover, these interim combinations are likely to serve as differentially effective retrieval cues depending on recent experience and, across individuals, as a function of more long-standing knowledge structures. The important point is that the underlying cognitive processes contributing to creative tasks remain relatively invariant, but, the contextual effects and how these influences interact with individual experience may determine what is retrieved, how that information is combined, and the choices of retrieval products to be retained or discarded. The last of these influences—namely, selection processes—is considered in detail next.

Selection Processes

Although generation and synthesis processes can provide important insights into how elemental components are retrieved and combined, they alone cannot explain which interim products will be retained as a part of the final product and which will be rejected. If a more complete cognitive account of creative behavior is to be offered, the cognitive mechanisms associated with the selection of information must be clearly identified. Unfortunately, few, if any, empirical studies have directly addressed the cognitive processes associated with such activities. However, in this section we present evidence for

possible cognitive mechanisms that specify how information might be selected for further elaboration and refinement in certain creative activities. As in previous sections, evidence is culled both from the existing literature on noncreative cognition and from the literature more directly focused on creative activity. The goal is to demonstrate that regularities exist in selection processes, just as regularities exist in generation and synthesis activities. We consider two mechanisms of selection specific to creative behavior and follow this with a more general consideration of cognitive selection processes that are likely to be operative during creative activities.

Selection Mechanisms Specific to Creative Behavior

The notion that creativity is based, in part, on the selection of information is not unique to the present framework. For example, one particular individual-attributes theory of creativity highlights the importance of selection mechanisms. Sternberg and Lubart's (1991, 1992, 1995) investment theory of creativity describes how some individuals have better intuition about selecting ideas that have greater potential to yield high impact, novel products. The theory is based on the simple economic principle of buying low and selling high. As applied to creative activity, one "buys low" when she or he adopts unknown or out-of-favor ideas that are perceived to have the potential to form a novel product. "Selling high" entails moving on to a new product, idea, or activity after the previous success has been used as a valuable contribution. According to the theory, there are six personal resources that allow an individual to efficiently select ideas that have value: intellectual abilities, knowledge, styles of thinking, personality, motivation, and environment. Creative individuals possess certain quantities of these resources, and other, less creative individuals possess them to a lesser degree or apply them in a less consistent fashion. For example, creative individuals use their prior knowledge to interpret the factors that might determine the ultimate value of a novel product. In addition, creative individuals expose themselves to a variety of ideas and have the ability to use them selectively.

Although those who favor an individual-

differences approach to creativity would be tempted to label the application of such strategies as a unique ability, our thesis is that such strategies basically constitute different cognitive agendas. By agenda, we mean nothing more than the constellation of cognitive processes that are used in approaching or solving a particular task, along with the degree to which they are engaged. For example, whereas one individual might choose a particular cognitive agenda or strategy in service of engaging in a creative task, another might choose an entirely different approach. Such mental agendas would either limit or expand the types of generation and synthesis processes exploited to complete the task. As different cognitive agendas are adopted, different types of information are generated, thereby producing variation among what information is selected for further refinement and what is abandoned as less relevant to the problem at hand. Mental agendas, and hence information brought to mind (to be subsequently refined), certainly differ across individuals but can also differ within individuals over time. Even the most creative people are uncreative at times, and the most uncreative person can offer the occasional creative insight. Therefore, creative behavior can be altered by idiosyncratic selections from among the elemental pieces of information made available by generative processes as a consequence of the mental agenda adopted.

Although investment theory does not highlight the cognitive mechanisms involved in creative processes, it does suggest very strongly that there are differences among individuals in how creative and noncreative endeavors are approached. Our point is that such differences may be largely attributed to strategic approaches that evaluate elemental information and the tentative combinations of that information. The important point is that investment theory illustrates how selection of information can be pivotal in the process of creative behavior. This theory also maintains that creative individuals possess knowledge structures (e.g., mental models) that help to define ultimate products as novel or creative. Therefore, the theory is consistent with the current framework insofar as prior knowledge will affect generation processes as well as selection of information based on that prior experience. In other words, investment theory essentially asserts that creative individu-

als use their knowledge structures (e.g., metaknowledge about creativity) as a basis for selecting among information that may be more or less appropriate for the particular task at hand.

Selecting From Available Information

These same selection principles contained in investment theory (as broadly construed here) can also be found in a more cognitively based theory of creativity. As mentioned earlier, the Genevieve model (Finke et al., 1992) posited two types of cognitive processing involved in creative activities: generation and exploration. For the current purposes, the relevant component of that theory to selection mechanisms is exploration. The goal of exploratory processes is to verify that preinventive structures produced by generative processes will lead to a successful resolution of the task ("resolution" depending on both the creative domain and the constraints imposed by task goals). The exploratory processes in Genevieve require that emergent properties of preinventive structures be interpreted in meaningful ways. There are a number of exploratory processes, and the particular type used is determined by the type of preinventive structure and the context of the task. However, each type is involved with the selection of preinventive properties that match the goals of the task at hand. Once selected, the preinventive structure is refined by cycling between generation and further exploration. For example, if the task is to create a new shape for an automobile, the designer might first generate a mental image of a novel shape for the vehicle. Then, during the exploratory process, the designer would select the properties of this preinventive form that make the shape more aerodynamic, allow for the maximum amount of passenger space, or involve some other attribute that is viewed as a novel contribution (i.e., an improvement over previous designs). These goals may require the generation of new shapes, further exploration, and so on.

Clearly, investment theory and Genevieve both imply that general knowledge structures (e.g., metaknowledge of creative tasks) guide the selection process of what information is appropriate to complete the task. However, the mental representations of the product must also be compared with the mental representations of

task goals and other cognitive agendas. In short, efficient selection requires the additional process of monitoring the evolving knowledge structures or mental representations. In Genevieve, generation continues until the task goals are completed or at least partially met. Therefore, the task goals must be monitored by additional cognitive processes so that, as generation and exploration continue, satisfactory refinement of the interim product can occur in stages toward a final form. In general, monitoring and selection processes constrain the product to match the task goals. Monitoring processes are obviously an extension of those processes that guide selection (or rejection) of task-relevant information.

Source Monitoring and Selection

As we have demonstrated throughout, the cognitive mechanisms used in the monitoring and selection process in creative activities have generally analogous counterparts that have been studied more extensively in the noncreative cognition literature. In particular, source monitoring defines one set of processes that can provide efficient mechanisms of novelty monitoring. At a general level, source monitoring involves the reconstruction of memory concerning the episodic origins of information. Therefore, if one is developing a new idea and can remember that it (or a similar idea) was encountered from a particular person at an earlier point in time, the "new" idea might not be viewed as very novel at all. The most complete model of source monitoring has been developed by Johnson and her colleagues (e.g., Johnson et al., 1993). According to this model, the qualitative characteristics of the encoding context (i.e., internal and external attributes of experience) are recorded in memory in association with the item or event itself. These characteristics are qualitatively different from one another and include perceptual detail, information about spatial and temporal attributes, affective information, and records of elaboration and encoding processes. Sufficient amounts of one or more of these qualitative characteristics are diagnostic to the source (i.e., origin) of information and hence can be used as a basis to determine whether an interim combination of elemental ideas or a potential

final product is truly novel as opposed to having an identifiable origin.

If the origin of information from the generation and synthesis stages can be identified as external to oneself as a result of source-monitoring-like selection processes, then that information should potentially be avoided as part of the creative end state because it may not be a product of one's own innovation. To our knowledge, there are no specific studies that have examined precisely how source monitoring is used in the service of creative activities. However, source-monitoring processes have been used extensively to explain other generative phenomena that are nevertheless related. For example, unconscious plagiarism or cryptomnesia (Reed, 1974; Taylor, 1965) can be described as an error in source monitoring. Unconscious plagiarism occurs when an individual offers information as novel when, in fact, the information was previously encountered from an external source (ignoring here the topic of self-plagiarism).

There are at least two ways in which source monitoring has been used to explain unconscious plagiarism. Each involves deviations from accurate source-monitoring processes. The strong implication of linking source monitoring with explanations of unconscious plagiarism is that memory for source can be used to prevent unconscious plagiarism except in those instances when source-monitoring processes are not properly engaged. One such instance is when source information is simply forgotten or not accessed when an item is brought to mind. For example, Brown and Halliday (1991) asked participants in small groups to generate category exemplars aloud. After various delays, the participants were again asked to generate "new" exemplars to the same categories. During the second generation attempt, participants wrote their responses individually and took memory tests for the original generations. Brown and Halliday found that increased rates of offering exemplars from the initial attempt as "new" (unconscious plagiarisms) coincided with a loss in source information over the delay period.

Although the parameters that influence source forgetting are not currently well defined (see Bornstein & LeCompte, 1995; Schacter, Harbluk, & McLachlan, 1984), there are factors that differentially affect retrieval of item information and source monitoring (Dodson & Johnson,

1996; Lindsay & Johnson, 1991). Therefore, it is logical that item information may be available to generative processes without the accompanying source information. In this case, the item (or idea) may come to mind readily as a result of recent exposure to the information. However, source information cannot be used to reject the generated information as "not novel," and so the generated information might be included in a novel product. Thus, unconscious plagiarism might occur as a result of dissociations between item and source information in the generation phase.

A second mechanism by which source-monitoring irregularities might contribute to the unconscious plagiarism phenomenon involves the failure to engage in source monitoring to a sufficient degree (Landau & Marsh, 1997; Marsh et al., 1997). Marsh and his colleagues used generative paradigms similar to those of Brown and Halliday (1991) with varying tasks (e.g., puzzle solutions and idea generation). However, the results of these studies showed unconscious plagiarism rates of about 20% even though source monitoring was quite good when it was explicitly tested. Marsh et al. interpreted this pattern to indicate that source information was available to the participants, but they did not use this information to monitor the novelty of their products during generation of new ideas. The implication is that unconscious plagiarism can arise from failures to monitor source information to a sufficient degree rather than from a loss of source information per se. Clearly, if people in Marsh et al.'s studies would have assessed the origin of generated information, they could have used their memory for the source to select against "old" information in their novel solutions (cf. Tenpenny, Keriakos, Lew, & Phelan, 1998).

The Metacognitive Model and Selection Processes

Up to this point, selection mechanisms have been characterized as those involved in determining the appropriateness (or inappropriateness) of available information for a creative task. The information evaluated by selection processes is made available through either generation processes or prior iterations of the creative process. Choosing from among available elemental pieces of information (and their interim recombina-

nations) represents only one function of selection processes as outlined in the present framework. Another important function that selection processes may serve is that of determining what generation (or possibly other selection) mechanisms are brought to bear in a given creative activity. The metacognitive model (Nelson, 1996; Nelson & Narens, 1994) can account for both functions of selection (i.e., the selection of information and the selection of relevant cognitive processes) in the present framework as well as illustrate how generation processes and selection processes interact.

The metacognitive model has not been a primary theoretical foundation for research on creativity but could provide the basis for mechanisms that both guide and monitor the selection of information in creative tasks. According to this model, cognitive processing for a given task takes place at different hierarchical levels of consciousness. The base level or object level represents conscious, task-driven processing of information. Higher order, or meta-level, processing monitors the output or processing of the object level. The meta-level processes also control the object-level processing through agenda selection, processing selection, and termination rules. Importantly, these two levels of cognition do not have "access" to the nature of the processing that is occurring at the other level. Each has access only to the flow of information from one level to the next (i.e., output). Monitoring represents the flow of information from object to meta level, and control represents the flow of information from meta to object level.

In the present framework, these mechanisms of metacognitive control are important for the consideration of process selection. Metacognitive control selects retrieval and execution strategies based on task demands, personal agendas, and the influence of metacognitive monitoring processes. In creative activity, the combination of processes selected to develop a product might be unique to the task at hand and might not yield the most efficient processing of information. For example, the study on unconscious plagiarism conducted by Marsh et al. (1997) demonstrated that highly differentiated source information was available to both creative (idea generation) and noncreative (source monitoring) tasks. This information was used accurately to provide source-monitoring

decisions, but the same memory for source should have indicated that the information was inappropriate for novel idea generation. It could be argued that processes were not brought to bear that could have avoided less-than-original contributions. Although this is true, obviously those processes were not seen by the creator as directly relevant to solving the task at hand. In the present framework, selection processes did not recognize the relevance or familiarity of elemental pieces of information (or interim combinations) and therefore did not specify that additional memorial searches were necessary to verify their novelty according to the current task demands. In this case, the information available at the object level did not invoke at the meta level a need to engage in further, more elaborated inspection processes. Because meta-level processes are responsible for such strategic selection of further processing, the failure to engage these processes is likely to be the result of task agendas that do not include "novelty" as a primary criterion in assessing whether a goal state has been achieved. Whether or not novelty is a component of people's cognitive agenda for such tasks is a potential topic for future empirical research.

This conclusion may be considered by some as rather strong given that traditional theories of creativity place a premium on the evaluation of final products vis-à-vis certain evaluative standards (i.e., product-centered theories of creative behavior). However, according to the present framework, this result should not be surprising at all. The processes by which a creator determines that the features of a product constitute a finished product are not the same processes by which an external party evaluates that product. To continue the example from the previous paragraph, the creator may use only heuristic assessments of familiarity that would not invoke more detailed novelty inspections of the components used in the product. If the creator uses a different strategy, these elements can be avoided and selected against before a conclusion about the end state is reached. By contrast, the cognitive processes of evaluation by an external party may begin with just such an inspection (i.e., application of the cognitive processes that the creator failed to use in sufficient degree). The important point is that metacognitive influences of the mental agenda of the creator and the evaluator do not

necessarily coincide. The same is true of product designers who create beautiful inventions on paper that are impossible to use in people's homes (Norman, 1988, chap. 6).

Interaction of Generation and Selection Processes

As mentioned previously, the metacognitive model is also useful for understanding the interaction between generation and selection processes. At a general level, task goals and the domain of activity comprise the cues for an initial search of relevant information for solving the creative problem at hand. These processes best correspond to the object level in this metacognitive description. As relevant information is retrieved and synthesized into interim combinations, these components constitute the memorial cues for the iterative search of additional relevant pieces of information to move toward a goal or end state. By contrast, at the meta level, these pieces of information and their interim combinations are evaluated and selected as useful or not useful for creating a product. Once decisions concerning their retention (or especially their rejection) have been made, these decisions change the working components at the object level and alter further generation and synthesis processes. As argued earlier, the influence of meta-level selection processes will depend wholly on the mental agenda of the creator, the domain and expertise in that domain, and the ultimate goal of the activity. The influence of processes at the basic level will be determined by the efficiency of retrieval processes, the degree to which information comes to mind explicitly and implicitly, and the way that synthesized pieces of information serve as iterative retrieval cues at this level.

This description is hardly an exhaustive account of how selection processes interact with those responsible for generation and synthesis. In part, the metacognitive model is not well specified because, like Runco and Chand's (1995) description of creative endeavors, its foundation rests on a set of interacting cognitive processes that are not easily dissociated from one another in laboratory research. For example, we have described that generation (and synthesis) processes are guided by both conscious and unconscious influences. The same was true in our depiction of selection processes. If each set

of processes were guided by one or the other type of influence, existing techniques common in cognitive psychology could be used to tease apart their impact on creative behavior. Dividing attention is known to reduce the availability of conscious resources. As a consequence, if meta-level selection processes were, say, largely conscious, then manipulations such as divided attention could be used to predict specific changes in creative output. The same would be true, for example, if basic-level processes involved in generation and synthesis were largely influenced by unconscious processes. In this case, manipulations such as changing modalities or introducing greater or lesser amounts of time between the introduction of relevant information and commencement of a creative task could be predicted to change behavior in predictable ways. The fact that this metacognitive model currently makes specific predictions about laboratory manipulation does not undermine its usefulness in understanding creative behavior. As manipulations are developed that can specifically interfere with basic-level processes (such as certain drugs that impair memorial retrieval but leave higher order processes intact) or meta-level processes, these manipulations can be used to fruitfully test the claims made by the model.

Conclusion

The overarching goal of this article was to provide a foundation by which those interested in creativity can understand the basic cognitive mechanisms of generating novel products. Our approach was to capture the processes that have been previously argued to account for creative behavior and place them within a slightly different framework. As with previous theories, the framework proffered here was designed to account for phenomena that are widely cited in research on creativity. Thus, our contribution to these endeavors probably does not lie in the degree to which our framework differs in radical ways from those theories that came before (e.g., Finke et al., 1992; Runco & Chand, 1995). Rather, our contribution lies in identifying the underlying cognitive processes involved in creative endeavors and, therefore, making this approach more tractable to further empirical and theoretical work. In the process, one of our major claims was that the cognitive processes

involved in creative activities are not qualitatively different from those regular processes involved in the course of everyday cognition. By this account, cognitive processes that are associated with the generation of relevant information and selection from among the generated pieces of information are not fundamentally different from those studied by psychologists immersed in exploring "noncreative" activities.

Of course, to those who hold aspects of the creative endeavor akin to the ethereal, our somewhat reductionist approach may appear woefully short in terms of explaining what defines the difference between a creative genius and a less creative individual. As we argued earlier, the problem lies in what one defines as creative. For example, perhaps novelty and variation are the hallmarks that distinguish creative products from less successful endeavors. The novelty requirement introduces an incredible amount of ambiguity into the task for the creator. More specifically, the creator faces the problem of deciding what information is most relevant in moving from a blank slate toward an appropriate end state. In the framework suggested here, the creator must start by generating relevant information, synthesizing that information, and selecting from among that information. We do not deny that, in the context of a creative activity, generating relevant information may be broadly construed to yield more domain-dependent and domain-independent information than in less creative tasks. In this sense, the creator who has generated more potentially relevant information must also sift through more of that information (using selection processes). Although the creative individual may have better capacities to handle these ambiguities, the underlying cognitive processes are unlikely to be very different. Likewise, variation in creative products within and across individuals may be due to idiosyncratic differences in mental agendas (as described earlier) or to differences in selecting among information that is generated.

The approach outlined here does not shatter in an iconoclastic way the fact that individuals will vary in their abilities to produce more or less creative products. Nor does it dispute that valuable insights can be gained by studying individual differences in abilities. However, the approach does suggest that examining more

traditional variables from cognitive psychology may allow some insight into additional underpinnings of the creative process. Our thesis is that individual variation in the same cognitive processes that are used by the majority accounts for differences in novelty and variation among creative products. These variables include, but are not limited to, working memory capacity, speed of retrieval, perceptual fluency, activation of relevant concepts and inhibition of irrelevant ones, recollective ability, inspection of memories, and a host of other processes that are used in everyday cognition. Thus, the search for correlates of creative ability should include those fundamental cognitive processes that are traditionally measured in noncreative research endeavors.

This approach may serve to open new avenues of inquiry as the variables just mentioned are studied in the context of creative endeavors. Equally as important, research on the least-well-defined area of this framework, namely, metacognitive control and selection processes, should also serve to open interesting lines of research. As examples, judgments of learning and feeling of knowing ratings (see Nelson, 1996) are standard cognitive methods used to assess people's monitoring of information during learning. Similar judgments could be taken in a creativity task to indicate people's assessment of novelty for generated information. Likewise, the metacognitive literature has assessed individuals' study time allocation and termination of processing to understand when control mechanisms are operative and when they are not. It could be particularly important to study the factors that influence the time and termination of generative processing. To date, there have been few if any cognitive studies on how individuals terminate generative processing. The theory and methodology of metacognition may provide the impetus for such investigations.

In closing, the ideas put forward here are not entirely novel insofar as they echo faint calls from long ago and from those who have operated on the fringe of research concerning creative endeavors. These ideas are bold, however, in that the framework and the specified cognitive processes were meant to be specific enough that they could be empirically tractable in the laboratory. As important, they were not intended to represent a full compendium of the

relevant cognitive processes; rather, they were intended to be modified and augmented as research on this exceptionally important topic continues. This framework and these ideas have served exceedingly well to guide our own research (with an admittedly cognitive orientation) as we investigate the regular cognitive processes that might be involved in creative behavior. We hope that they will serve equally well for those with an admitted interest in creativity to speculate on additional cognitive processes that we have failed to identify here. Perhaps with such an iterative interplay between disciplines creative advances can be made in the understanding of creative behavior.

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